



ecoconnect

— CONNECTING TODAY, SUSTAINING TOMORROW —

— نربط اليوم، نستدام غداً —

UAE GOVERNMENT



حكومة دولة الإمارات العربية المتحدة

Tatweer Hackathon 2026

EcoConnect: The UAE Sustainability Super-App

Empowering Remote Communities | Aligned with
the Ministry of Climate Change & Environment

Omar Abdelfattah | Tatweer Hackathon 2026



Vision & Public Ecosystem

- A Decentralized Future
- Enabling a Trusted, Resilient Data Exchange Framework
- Integrated Services for Citizens
- Digital Identity Integration
- Smart City Optimization
- Sustainability as a Core Metric
- Real-time Insights and Feedback
- A Scalable, Open Architecture
- Citizen Empowerment through Data



ECOCONNECT

THE UAE SUSTAINABILITY SUPER-APP
EMPOWERING REMOTE COMMUNITIES | MINISTRY OF CLIMATE CHANGE & ENVIRONMENT
OMAR ABDELFAH | TATWEER HACKATHON 2026

VISION

Bridging UAE's remote agricultural communities — Hatta, Liwa, Al Dhafra — with urban centers through a unified Super-App. Consolidating fragmented services to empower farmers, attract eco-tourists, and secure green investments. Fully aligned with the Ministry of Climate Change & Environment goals.





PUBLIC ECOSYSTEM NO LOGIN REQUIRED

EcoConnect's public-facing platform offers open access to key sustainability tools, marketplaces, and interactive experiences — no account needed.

01

Landing Portal

Mission intro and platform overview — discover EcoConnect's vision, features, and alignment with UAE sustainability goals.

02

Eco Souq Marketplace

A vibrant e-commerce hub connecting buyers with local farmers' produce, handcrafted goods, and sustainable products from UAE's remote communities.

03

QR Traceability Engine

On-chain verification for product origin, water usage, and soil certificates — ensuring full transparency across the supply chain.

04

Vincent Mobile App Simulator

Twin-phone interactive demo: Vincent EcoConnect (AI Advisor, GPS logging, SOS Beacon) and Vincent Souq (Shop → Cart → Receipt consumer flow).



ROLE-BASED ARCHITECTURE

FARMER & ECO-TOURIST

Two core user experiences powering EcoConnect — from precision farm management to immersive eco-tourism, each role is purpose-built for impact.

01

FARMER: AGRI-BUSINESS HUB

Farm OS: Market & Trade (Eco Market, Tool Sharing), Operations & Finance (My Farm, Payroll, Subsidies), Sustainability (Eco-Grid, Eco Credits, Gov-Connect), Safety & Intelligence (SOS Beacon, Falcon AI Copilot).

02

ECO-TOURIST EXPERIENCE

Explore: Eco-Tourism bookings, Itinerary Builder, Local Eco-Events. Engage: Earn & spend Eco-Credits, Eco-Shield for SOS & incident reports, ensuring safe and sustainable travel.

ROLE-BASED ARCHITECTURE: INVESTOR, EMPLOYEE, SELLER & GOV ADMIN

EcoConnect delivers tailored experiences for four additional roles, each with dedicated tools for their specific needs.

01

Investor (Green FinTech)

Discover rural startup funding pipelines, run Market Intelligence Scans, and execute Smart Contract signature & receipt workflows.

02

Employee (Workforce Management)

Access task dashboards, timesheets, payslips, and clock-in tools. Collaborate via Kanban Task Board and integrated Team Chat.

04

Government Admin (Central Governance)

Oversee user and role management, issue instant permit approvals, and generate system-wide compliance reports.

03

Seller (E-Commerce Operations)

Manage product listings, track inventory, and process orders seamlessly through the integrated Eco Souq Marketplace.

CROSS-CUTTING TECHNICAL CAPABILITIES

The underlying technical features powering EcoConnect's unified Super-App experience across all user roles.

01

Intelligent AI Integration

Local LLM copilot, AI advisor, and market insight engine with rule-based fallbacks for reliable offline support.

02

Seamless Authentication & KYC

JWT login, secure role-based routing, and AI-powered document verification for trusted onboarding.

03

Automated Government Services

Real-time subsidy eligibility checks, auto-generated PDF certificates, and instant government reports.

04

Inclusive Design

Full bilingual support (English / العربية) with a dedicated accessibility panel for all users.

TECH STACK & CLOSING

Frontend: React 19 + Vite + Tailwind CSS | Backend: Python + FastAPI + SQLAlchemy + SQLite | Utilities: reportlab (PDF generation), On-chain traceability

EcoConnect unifies UAE's remote communities with urban centers through an intelligent Super-App.

Built by Omar Abdelfattah | Tatweer Hackathon 2026

Six Role-Based User



Farmer

- Market & Trade tools
- Operations & Finance
- Eco Grid tracking



Eco-Tourist

- Bookings & Itinerary
- Eco-Credits rewards
- Eco-Shield safety



Employee

- Real-time tasks & timesheets
- Team Chat collaboration
- Payslips & Kanban



Investor

- Funding pipeline
- Market Intelligence
- Smart Contract execution



Seller

- E-commerce tools
- Inventory management
- Eco Souq integration



Government Admin

- User & role management
- Instant approvals
- System reporting

Technical Architecture

AI Integration

Local LLM copilot, market insights,
rule-based fallbacks

Auth & KYC

JWT login, role-based routing,
AI document verification

Gov-Services

Real-time subsidy checks,
auto PDF certificates

Bilingual

English / Arabic +
accessibility panel

Stack

React 19 + Vite + Tailwind CSS
Python + FastAPI + SQLAlchemy + SQLite
reportlab PDF | On-chain traceability

Try It Live

Scan to access the platform



<http://127.0.0.1:8000/#/login>